The Five-Absence Predictions Set — BST's Simultaneous-Null Falsifier Framework (expanded to 7 entries 2026-06-29 via F397 Casey ratification)

Casey Koons (named principle) + Elie (Toys 3102/3103 + Five-Absence synthesis) + Keeper (formalizatio

2026-05-19 Tuesday late afternoon; expanded 2026-06-29 Monday with F397 NO 4th generation (Casey ratification per K596)

The Five-Absence Predictions Set (7 entries)

Name retained for historical continuity; count now SEVEN per Casey ratification 2026-06-29 (K596) admitting F397 NO 4th generation.

Casey-named statement (updated 2026-06-29)

> "BST predicts the simultaneous absence of seven conjectured-but-unobserved physics objects: no Grand Unified Theory, no proton decay, no Dark Matter particle, no magnetic monopoles, no sterile neutrinos, no supersymmetry, and no fourth fermion generation. ANY single positive detection in any of these seven channels refutes the entire BST framework. The simultaneous-null prediction cannot be evaded by parameter-adjusting — it is a structural commitment of D_IV^5 geometry."

The seven absences with BST mechanism

#	Absence	BST mechanism	Current experimental status	Falsifier threshold
1	No GUT	Gauge groups from DISTINCT D_IV^5 features: SU(3)←N_c=3 (color), SU(2)←rank=2 (symmetry), U(1)←c_2=11 (Chern). Never unify. The “10^16 GeV coupling convergence” is RG kinematic, not physical unification. Confirmed via Elie 2026-06-29: pairwise β-running crossings miss by ~10^4 in scale (10^13 / 2×10^14 / 10^17 GeV; GUT triangle does not close)	No GUT confirmed at any energy scale (40+ years null)	Discovery of fourth gauge boson at unification scale OR proton decay (which would imply GUT)
2	No proton decay (τ_p = ∞)	Proton = bulk geodesic on D_IV^5; complete N_c-phase commitment; topologically stable (3-component trefoil link, W-23). No GUT → no dim-6 operators → strict zero decay rate	Super-K limit τ_p > 1.6×10^{34} yr; Hyper-K + DUNE will push limits	ANY positive proton decay observation refutes BST
3	No DM particle	Dark Matter = geometric remainder, not new particle (per BST Saturday framework). DM is the “incomplete winding” residue accounting for cosmological observations	LZ + XENONnT + DARWIN extending direct-detection null limits; LHC no WIMP discovery	Direct WIMP detection at terrestrial detector refutes BST

#	Absence	BST mechanism	Current experimental status	Falsifier threshold
4	No magnetic monopoles	Substrate topology forbids isolated magnetic charge. No GUT-breaking → no Dirac-Polyakov-'tHooft monopoles	All searches null (MoEDAL, IceCube, etc.)	Positive monopole detection refutes BST
5	No sterile neutrinos	PMNS triangle complete (T1932); three-generation closure is structural (per W-20 mass hierarchy from Wallach layer index). No fourth generation possible	MicroBooNE etc. null; no sterile evidence	Discovery of fourth-generation neutrino refutes BST
6	No supersymmetry	BST has no SUSY structure; superpartners do not exist. The cancellations SUSY was invented to provide come from D_IV^5 Bergman geometry directly. Confirmed via Elie 2026-06-29: SUSY was historically invented to force GUT-triangle closure; BST forbids SUSY → coupling crossings must miss → confirmed	LHC null through Run 3; HL-LHC null projected	Discovery of any superpartner refutes BST

#	Absence	BST mechanism	Current experimental status	Falsifier threshold
7	No fourth fermion generation (NEW 2026-06-29)	F86 Korányi-Wolf strata rigidity (theorem-level): rank-2 bounded symmetric domain D_IV^5 has EXACTLY rank+1 = 3 boundary strata; no fourth stratum possible. Per Cal #453 precision rider: source is F86 strata rigidity (theorem), NOT Wallach set (which has a continuum with 2 discrete points). F395 Lyra+Grace Wallach/F86 dual characterization identifies generations with strata but only F86's rigidity forbids 4th.	LEP confirmed: 3 light neutrino species via Z width measurement (1989); no 4th-generation charged lepton, quark, or neutrino observed	Discovery of 4th-generation charged lepton, quark, or neutrino refutes BST

Why this is BST's strongest external argument

Standard physics frameworks (SUSY, GUT, seesaw, WIMP DM) all predict positive detections somewhere. Parameter-adjusting can move predicted signatures around in mass/coupling space but predicts SOMETHING positive.

BST predicts simultaneous null across all six channels. There is no parameter-adjustment that maintains BST consistency while predicting any of these positives. The framework either holds across all six OR fails.

This is the structural risk-taking that distinguishes BST from speculation. Six independent falsifier channels, each with experimental programs underway.

Five-absence framework as one structural commitment

Critical honest framing: the six absences are NOT six independent predictions. They are six consequences of one underlying BST framework:

- Distinct D_IV^5 features (not unified) → no GUT
- Bulk-geodesic stability + no GUT → no proton decay
- Geometric remainder accounting → no DM particle
- Substrate topology + no GUT-breaking → no magnetic monopoles
- Closed three-generation structure → no sterile neutrinos
- D_IV^5 Bergman geometry handles cancellations → no SUSY

This is theoretical economy (six predictions from one framework) NOT five-fold independent confirmation. Cal coincidence-filter Mode 7 (single-mechanism over-claim) is the honest discipline that prevents over-counting these as independent predictions.

External presentation must specify: simultaneous null comes from ONE framework's structural commitments; falsifying ANY ONE absence falsifies the framework's underlying structural commitment (and hence the other five predictions).

Tier discipline (T2385 split-tier)

Level 1 (D-tier framework status, per Grace INV-4474): - Each absence has documented BST mechanism from established D_IV^5 structure - All six absences currently consistent with experiment (40+ years null) - Falsifier specification is concrete and falsifiable (positive detection thresholds)

Level 2 (I-tier interpretation): - "Cannot be evaded by parameter-adjusting" — needs Cal coincidence-filter ratification - "Structurally stronger than positive predictions" — comparative claim about epistemic strength - "BST's strongest external argument" — ranking claim against other BST results

The principle holds at D-tier framework level. The "strongest argument" comparative claim is I-tier interpretation.

Pairing with proton over-determination

The Five-Absence Set is most powerful when paired with the proton over-determination headline:

#	Proton observable	Tier	BST form
1	m_p/m_e	D, 0.002%	$6\pi^5$
2	Proton radius	D, 0.02%	rank$^2\cdot\hbar$c/m_p
3	μ_n/μ_p	D, exact	137/200
4	g-factor	I, 3.5 ppm	391/70
5	Lifetime	D, null	∞

Combined external headline: *"BST predicts five proton observables (all D/I-tier or absolute null) AND six universal absences (GUT, proton decay, DM particle, monopoles, sterile neutrinos, SUSY). All currently consistent with experiment. Single positive detection in any absence channel refutes the entire framework. Single failed proton observable in next-generation experiments refutes the framework. The framework is over-constrained, not over-fit."*

Falsifier specification

The Five-Absence Set falsifies under ANY of:

1. Positive proton decay (τ_p $< \infty$)
2. Discovery of GUT gauge boson at unification scale
3. Direct WIMP detection (LZ, XENONnT, DARWIN, etc.)
4. Positive magnetic monopole detection
5. Discovery of fourth-generation neutrino
6. Discovery of any supersymmetric partner

ANY ONE positive observation in ANY ONE channel falsifies BST entirely.

Connection to Substrate Working Process Principle

The Five-Absence Set and SWPP are companion frameworks: - SWPP: substrate as active working process (positive ontology) - Five-Absence Set: what the substrate does NOT contain (negative ontology)

Together they specify BST's complete predictive shape: positive observable structure (via SWPP commitments) + null channels (via Five-Absence Set). Standard physics frameworks usually have one or the other; BST has both with explicit falsifiers for each.

K-audit status

K64 candidate (Keeper queue): ratify Five-Absence Framework against Cal coincidence-filter Modes 1 and 7, plus Casey-named principle status. Pre-audit verdict: framework is structurally sound, well-disciplined regarding theoretical-economy framing.

Per Casey's standard

- Simple: six things BST says do not exist. Each has a mechanism. Each has a falsifier.
- Works: 40+ years of null observations are consistent with the predictions
- Hard to break: cannot be evaded by parameter-adjusting because all six emerge from one structural framework
- Counter-example: positive detection in any one channel breaks the framework — none observed

Operational targets

The Five-Absence Set defines what successful BST external presentation looks like:

1. State the simultaneous-null framework with all six channels
2. Mechanism each absence from BST structural commitments
3. Threshold each falsifier explicitly
4. Frame as theoretical economy (one framework, six consequences) not five-fold independence
5. Pair with proton over-determination for combined headline
6. Acknowledge the baryogenesis honest gap alongside (per Casey's Saturday discipline)

External 1-pager work queued (Option B per Casey)

- Lyra: draft external 1-pager based on this principle + proton over-determination
- Cal: gate-pass for external survivability (Mode 1, Mode 7 specifically)
- Grace: external survivability check + reference table
- Keeper: K64 audit when 1-pager reaches v0.2

— Filed by Keeper, 2026-05-19 Tuesday EDT, per Casey directive "Option B for external + Option A for internal"